

Which Rewards Matter? A Controlled Component Ablation of GRPO for Small-Model Mathematical Reasoning

Moh. Izzatul Haqi* [TODO: Advisor name (co-author)]
[TODO: affiliation / lab]

Abstract

Reinforcement learning with verifiable rewards (RLVR) has become a standard recipe for eliciting mathematical reasoning in language models, and the reward functions used in practice increasingly combine several signals—answer correctness, output format, length control, and partial-credit terms. Yet the *marginal* contribution of each component is rarely isolated, and results are frequently reported from a single training run, leaving open which signals genuinely help and which merely invite reward hacking. We present a controlled, multi-seed ablation of the reward function in Group Relative Policy Optimization (GRPO) on a small (1.5B-parameter) model trained under a fixed QLoRA budget. We decompose the reward into correctness, format, length, and partial-credit components and evaluate a cumulative ladder of five variants (A0–A4) on GSM8K and on out-of-distribution MATH-500, reporting paired significance tests and effect sizes across seeds. We find that [TODO: key finding 1 (e.g. which component yields the largest pass@1 gain)], that [TODO: key finding 2 (e.g. a component that helps the reward signal but not held-out accuracy)], and that [TODO: key finding 3 (e.g. the pass@1 vs. pass@k pattern)]. Our results [TODO: one-sentence takeaway]. Code is available at <https://github.com/mohizzatulhaqi/grpo-reward-ablation>.

1 Introduction

Reinforcement learning has emerged as the dominant approach for improving the reasoning ability of large language models (LLMs). Where early work relied on learned reward models trained from human preferences [8], recent systems for mathematical and programmatic reasoning instead use *verifiable* rewards: a rule checks whether the final answer is correct, yielding a sparse but unambiguous training signal [4]. Group Relative Policy Optimization (GRPO) [11] has made this recipe particularly practical by estimating advantages from a group of sampled completions, removing the need for a separate value network and substantially lowering the cost of RL fine-tuning.

In practice, the reward used to train reasoning models is rarely a single term. Implementations commonly add a *format* reward to encourage parseable output, a *length* penalty to curb verbosity, and various *partial-credit* or process signals intended to densify the otherwise sparse, end-of-sequence reward and improve credit assignment [7]. Each addition is plausible in isolation, but the components interact, and a poorly designed term can be exploited: a model may learn to satisfy a proxy (e.g. produce a well-formatted or short answer) without improving its actual accuracy. Despite the centrality of these design choices, their individual contributions are seldom measured in a controlled

*Department of Information Technology, Institut Teknologi Sepuluh Nopember (ITS), Surabaya, Indonesia. Correspondence: <email>.

way, and the RLVR literature frequently reports single-seed results, making it difficult to separate genuine effects from run-to-run variance.

This paper asks a deliberately narrow but practically important question: *within a fixed GRPO setup, which reward components actually improve reasoning, which are redundant, and which induce reward hacking?* We answer it with a cumulative ablation on a small (1.5B) model. Starting from a pure correctness reward (RLVR), we add one component at a time—format, length, and partial credit—defining five variants A0 through A4. Crucially, we hold the optimization budget fixed (the same QLoRA configuration, training steps, and group size across variants) so that differences are attributable to the reward, and we run each variant across multiple random seeds, reporting paired significance tests and effect sizes rather than point estimates.

We deliberately study the small-model, limited-compute regime. This setting is both increasingly relevant—efficient post-training of small open models is a practical concern—and methodologically convenient: a general-purpose (rather than math-specialized) base model leaves headroom for the reward to move accuracy, and an out-of-distribution evaluation on MATH-500 tests whether gains generalize beyond the training distribution.

Contributions.

- We formulate a controlled, cumulative reward ablation (A0–A4) for GRPO and instantiate it with four reward components whose definitions are given in Section 3.
- We adopt a statistical protocol—multiple seeds with paired Wilcoxon signed-rank tests and Cliff’s δ effect sizes—that is rarely applied in the RLVR setting, and we report a reward-hacking diagnostic that tracks each component’s reward against held-out accuracy.
- Empirically, we find that **[TODO: summarize 2–3 findings once results are in]**. All code and configurations are released for reproducibility.

2 Related Work

RL for LLM reasoning. Reinforcement learning from human feedback established RL as a tool for aligning LLM behaviour [8], typically optimized with Proximal Policy Optimization (PPO) [10]. For tasks with checkable answers, RL with verifiable rewards replaces the learned reward model with a deterministic correctness check, which DeepSeek-R1 used to elicit strong reasoning without supervised reasoning traces [4]. GRPO [11] simplifies the optimizer by computing group-relative advantages, and is now a common choice for reasoning-oriented fine-tuning.

Reward design and credit assignment. Because the verifiable reward is sparse—granted only at the end of a long generation—a recurring theme is densifying or shaping it. Process-supervised rewards score intermediate steps and can outperform outcome-only supervision on mathematical reasoning [7]. Format and length terms are widely used in practice to stabilize training and control output, but they are proxies whose effect on true accuracy is not guaranteed. Our partial-credit component is a lightweight, rule-based instance of this densification idea, and our ablation is precisely an attempt to quantify when such shaping helps.

Does RL expand or sharpen reasoning? A active debate concerns whether RLVR expands a model’s reasoning capability or merely concentrates probability on solutions the base model could already sample. Evidence that RL improves pass@1 while leaving pass@ k (for large k) largely

unchanged suggests a sharpening rather than expansion effect [13]. We connect to this question directly by reporting pass@ k alongside pass@1 for every variant.

Efficient post-training. Parameter-efficient methods make this study feasible on a single consumer GPU. LoRA adapts a frozen model with low-rank updates [6], and QLoRA [3] further quantizes the base model to 4-bit, enabling RL fine-tuning of billion-parameter models in modest memory. We use QLoRA uniformly across all variants; because the comparison is *relative*, a fixed parameter-efficient setup does not confound the ablation.

3 Method

3.1 Background: GRPO

For a prompt q , GRPO samples a group of G completions $\{o_1, \dots, o_G\}$ from the current policy $\pi_{\theta_{\text{old}}}$ and scores each with a scalar reward $r_i = R(o_i)$. Advantages are estimated by normalizing rewards *within* the group,

$$\hat{A}_i = \frac{r_i - \text{mean}(\{r_1, \dots, r_G\})}{\text{std}(\{r_1, \dots, r_G\}) + \varepsilon}, \quad (1)$$

which removes the need for a learned value function. The policy is then updated with a clipped objective and a KL penalty toward a reference policy π_{ref} :

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min(\rho_{i,t} \hat{A}_i, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) \hat{A}_i) - \beta \text{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right], \quad (2)$$

where $\rho_{i,t} = \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})$ is the per-token importance ratio.

3.2 Reward components

We decompose the reward into four components, each scaled to $[0, 1]$ so that the weights below act as clean relative importances. Let $\text{ext}(o)$ denote the extracted final answer of completion o , a^* the gold answer, and $\mathcal{N}(\cdot)$ the multiset of numbers appearing in a string.

- **Correctness** (the only term in A0): $r_{\text{corr}}(o) = \mathbb{1}[\text{ext}(o) = a^*]$.
- **Format** (structure, independent of value): $r_{\text{fmt}}(o) = \frac{1}{2} \mathbb{1}[\text{reasoning block present}] + \frac{1}{2} \mathbb{1}[\text{parseable boxed answer}]$.
- **Length** (soft brevity, budget T , hard cut H): $r_{\text{len}}(o) = \text{clip}\left(1 - \frac{\max(0, |o| - T)}{H - T}, 0, 1\right)$.
- **Partial credit** (process signal via intermediate-number overlap with the gold solution g):

$$r_{\text{part}}(o) = \frac{|\mathcal{N}(o) \cap \mathcal{N}(g)|}{|\mathcal{N}(g)|}.$$

The full reward is a weighted sum $R(o) = \sum_k w_k r_k(o)$ over the active components of a given variant.

Table 1: Reward components active in each ablation variant. Weights are starting values tuned on a single-seed sweep; *report final A4 weights*.

Variant	Correctness	Format	Length	Partial	Description
A0	✓				pure RLVR baseline
A1	✓	✓			+ format
A2	✓	✓	✓		+ length control
A3	✓	✓	✓	✓	+ partial credit
A4	✓	✓	✓	✓	full, tuned weights

Algorithm 1 GRPO training with composite reward (one variant)

Require: base policy π_θ , dataset $\mathcal{D}$, group size G , active components and weights $\{w_k\}$

- 1: **for** each training step **do**
- 2: sample prompt $q \sim \mathcal{D}$
- 3: sample group $\{o_1, \dots, o_G\} \sim \pi_{\theta_{\text{old}}}(\cdot \mid q)$
- 4: **for** $i = 1$ to G **do**
- 5: $r_i \leftarrow \sum_k w_k r_k(o_i)$ ▷ composite reward
- 6: **end for**
- 7: $\hat{A}_i \leftarrow (r_i - \text{mean}(r)) / (\text{std}(r) + \varepsilon)$
- 8: update θ by minimizing $\mathcal{L}_{\text{GRPO}}$
- 9: **end for**

3.3 Ablation design

The five variants form a cumulative ladder (Table 1): each adds one component to the previous. A0 is pure RLVR and serves as the internal baseline against which all others are compared; A4 uses the full reward with weights tuned on a single-seed sweep. Algorithm 1 summarizes the training loop. The format and correctness terms are intentionally orthogonal—a completion can be well-formatted yet wrong—so that “follows the format” and “gets the right answer” can be measured separately.

4 Experimental Setup

Models. Our primary model is Qwen2.5-1.5B-Instruct [9]; we deliberately use the general instruction-tuned model rather than the math-specialized variant so that the reward has room to change accuracy. Pipeline prototyping uses Qwen2.5-0.5B-Instruct.

Data. We train on GSM8K [2] (7,473 grade-school math problems), whose answers are trivially verifiable. We evaluate on the GSM8K test split (1,319 problems) and, as an out-of-distribution test, on MATH-500 [7], a 500-problem subset of the MATH dataset [5].

Training. All variants are trained with GRPO, implemented in the TRL library [12], under an identical budget: 4-bit NF4 quantization with LoRA (rank 16, $\alpha = 32$) applied to all attention and MLP projections [6, 3], group size $G = 8$, KL coefficient $\beta = 0.04$, learning rate $1e-6$, and N optimization steps, with gradient checkpointing on a single 16 GB GPU. The KL reference is obtained by disabling the LoRA adapter, so no second full-size model is instantiated.

Table 2: Main results (mean over seeds). *All numbers are placeholders.*

Variant	pass@1		pass@ n	mean len	format rate
	GSM8K	MATH-500			
A0	-.-	-.-	-.-	—	-.-
A1	-.-	-.-	-.-	—	-.-
A2	-.-	-.-	-.-	—	-.-
A3	-.-	-.-	-.-	—	-.-
A4	-.-	-.-	-.-	—	-.-

Table 3: Comparison to the A0 baseline on GSM8K pass@1: difference, paired Wilcoxon p , and Cliff’s δ . *Placeholders.*

Variant	Δ vs. A0	p (Wilcoxon)	Cliff’s δ
A1	+-.-	-.-	+-.-
A2	+-.-	-.-	+-.-
A3	+-.-	-.-	+-.-
A4	+-.-	-.-	+-.-

Evaluation. For each test problem we sample $n = 8$ completions at temperature 0.8. We report pass@1 (the unbiased estimator c/n averaged over problems) and pass@ n using the estimator of Chen et al. [1], together with mean completion length and the rate of format adherence.

Statistical protocol. Each variant is trained with 3 random seeds. We report mean $\pm$ standard deviation and compare each variant to A0 with a paired Wilcoxon signed-rank test (seeds matched across variants) and Cliff’s δ effect size. We note that with 3 seeds the smallest attainable paired p -value is bounded away from conventional thresholds; we therefore foreground effect sizes and treat p -values as supporting evidence rather than the primary claim.

Reproducibility. Reward functions, the verifier, the statistical analysis, and the resume-capable training driver are released, with unit tests covering the components that determine the reward signal and the reported metrics.

5 Results

[**TODO:** Replace the placeholder numbers in Tables 2 and 3 with aggregated results, and add learning-curve and accuracy–length figures.]

Per-component contribution. [**TODO:** Report the cumulative ladder trend and (if run) a leave-one-out on A4 that drops each component to measure its marginal effect, especially partial credit.]

Reward hacking. [**TODO:** Present the diagnostic: for each variant, contrast the trajectory of each component’s reward with held-out accuracy; flag any component whose reward rises while accuracy stalls (e.g. the length term incentivizing short wrong answers).]

6 Analysis and Discussion

[**TODO:** Structure this around the questions the design was built to answer:]

- [**TODO:** Which components produced statistically/effect-size-meaningful gains, and which did not?]
- [**TODO:** pass@1 vs. pass@ n : does the reward shaping expand reasoning or sharpen the existing distribution [13]?]
- [**TODO:** The length-reward trade-off (unconditional vs. correctness-gated), if the sub-experiment was run.]

Limitations. This study uses 3 seeds, a single model size, a single task family (grade-school and competition math), and a parameter-efficient (QLoRA) rather than full fine-tuning setup; effect sizes are therefore more reliable than p -values, and replication at larger scale and with full fine-tuning is left to future work. The partial-credit signal is intentionally simple (number overlap) and is not a learned process-reward model.

7 Conclusion

We presented a controlled, multi-seed ablation isolating the contribution of individual reward components in GRPO for small-model mathematical reasoning. [**TODO:** State the headline finding and its practical implication for reward design in one or two sentences.]

References

- [1] Mark Chen et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.
- [2] Karl Cobbe et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- [3] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized LLMs. *arXiv preprint arXiv:2305.14314*, 2023.
- [4] Daya Guo et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- [5] Dan Hendrycks et al. Measuring mathematical problem solving with the MATH dataset. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks*, 2021.
- [6] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- [7] Hunter Lightman et al. Let’s verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.
- [8] Long Ouyang et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

- [9] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- [10] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- [11] Zhihong Shao et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- [12] Leandro von Werra et al. TRL: Transformer reinforcement learning. <https://github.com/huggingface/trl>, 2020.
- [13] Yang Yue et al. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? *arXiv preprint arXiv:2504.13837*, 2025.